INTERNSHIP REPORT

Epidemiological Data Collection and Analysis of Vector-Borne and Viral Hemorrhagic Fever Outbreaks: Ebola (Democratic Republic of Congo), Dengue (Multi-Country), and Chikungunya (Madagascar)

Submitted in partial fulfilment of the requirements for the internship program

Submitted by

Supriyo Saha

4-year B.Sc. Honours in Statistics

Maulana Azad College

Acknowledgement

I would like to express my sincere gratitude to Prof. Sheikh Taslim Ali of Hong Kong University for the opportunity to intern under his guidance and for his continuous support and mentorship throughout this internship.

Abstract

This report summarizes the work carried out during a research internship focused on the collection, organization, and visualization of infectious disease surveillance data. The internship comprised three components: (1) collection of Ebola virus disease case data for the Democratic Republic of Congo relating to the 17th Ebola epidemic (2025–2026); (2) collection and visualization of dengue case data across multiple countries, including Bangladesh, Brazil, Colombia, Hong Kong, Malaysia, Mexico, Peru, the Philippines, Singapore, and Thailand, using R and RStudio; and (3) collection of chikungunya case data for Madagascar for the year 2025. The report describes the methodology, tools, and outputs associated with each task.

1. Introduction

Vector-borne and viral hemorrhagic fever diseases such as Ebola, dengue, and chikungunya remain significant public health concerns globally. Accurate and up-to-date case data is essential for understanding outbreak dynamics, seasonal patterns, and geographic spread. This internship was undertaken under the guidance of Prof. Sheikh Taslim Ali from Hong Kong University, with the objective of supporting ongoing epidemiological research through structured data collection and visualization.

2. Objectives of the Internship

- To collect and organize case data for the 17th Ebola epidemic in the Democratic Republic of Congo (2025–2026).
- To collect dengue case data for multiple countries and produce time-series visualizations of monthly/yearly case counts using R.
- To set up and use R and RStudio as a working environment for epidemiological data analysis.
- To collect chikungunya case data for Madagascar for the year 2025.
- To develop practical skills in data cleaning, reproducible analysis, and scientific reporting.

3. Tools and Environment

The following tools were used during the internship:

- Base R — statistical computing environment
- RStudio — integrated development environment (IDE) for R
- R packages: tidyr, dplyr, ggplot2, readr
- Microsoft Excel / CSV files for raw data organization
- Source of data: Ebola OLD Situation Report (SITREP) (https://insp.cd/ebola/)
- Dropbox – Data storage, file organization, and sharing

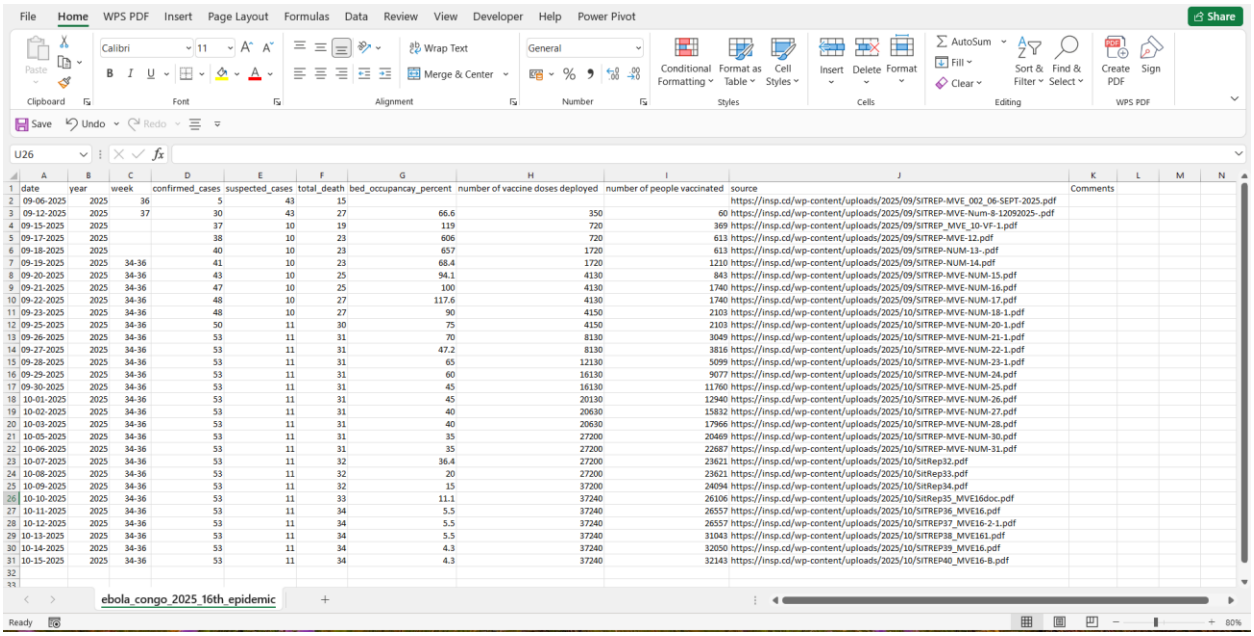
4. Work Undertaken

4.1 Ebola Case Data Collection — Democratic Republic of Congo (16th Epidemic, 2025)

Data on confirmed, probable, and suspected Ebola virus disease cases were collected for the ongoing 17th epidemic in the Democratic Republic of Congo. The data collection covered the period 2025–2026 and involved compiling figures from publicly available situation reports and health authority bulletins.

Here, we collect the following variables: [Date, Year, Week, Confirmed_Cases, Suspected_Cases, Total_Deaths, Bed_Occupancy_Present, Number_of_Vaccines_deployed, Number_of_people_get_vaccinated.]

Also, here I have faced problems like some information is missing in the source. Then it is very time-consuming, and we are not sure that the collected data is true.



date	year	week	confirmed_cases	suspected_cases	total_death	bed_occupancay_percent	number of vaccine doses deployed	number of people vaccinated	source	Comments
09-06-2025	2025	36	5	43	15				https://insp.cd/wp-content/uploads/2025/09/SITREP-MVE_002_06-SEPT-2025.pdf	
09-12-2025	2025	37	30	43	27	66.6	350	60	https://insp.cd/wp-content/uploads/2025/09/SITREP-MVE-Num-8-12092025-.pdf	
09-15-2025	2025		37	10	19	119	720	369	https://insp.cd/wp-content/uploads/2025/09/SITREP_MVE_10-VF-1.pdf	
09-17-2025	2025		38	10	23	606	720	613	https://insp.cd/wp-content/uploads/2025/09/SITREP-MVE-12.pdf	
09-18-2025	2025		40	10	23	657	1720	613	https://insp.cd/wp-content/uploads/2025/09/SITREP-NUM-13-.pdf	
09-19-2025	2025	34-36	41	10	23	68.4	1720	1210	https://insp.cd/wp-content/uploads/2025/09/SITREP-NUM-14.pdf	
09-20-2025	2025	34-36	43	10	25	94.1	4130	843	https://insp.cd/wp-content/uploads/2025/09/SITREP-MVE-NUM-15.pdf	
09-21-2025	2025	34-36	47	10	25	100	4130	1740	https://insp.cd/wp-content/uploads/2025/09/SITREP-MVE-NUM-16.pdf	
09-22-2025	2025	34-36	48	10	27	117.6	4130	1740	https://insp.cd/wp-content/uploads/2025/09/SITREP-MVE-NUM-17.pdf	
09-23-2025	2025	34-36	48	10	27	90	4150	2103	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-18-1.pdf	
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09-26-2025	2025	34-36	53	11	31	70	8130	3049	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-21-1.pdf	
09-27-2025	2025	34-36	53	11	31	47.2	8130	3816	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-22-1.pdf	
09-28-2025	2025	34-36	53	11	31	65	12130	5099	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-23-1.pdf	
09-29-2025	2025	34-36	53	11	31	60	16130	9077	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-24.pdf	
09-30-2025	2025	34-36	53	11	31	45	16130	11760	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-25.pdf	
10-01-2025	2025	34-36	53	11	31	45	20130	12940	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-26.pdf	
10-02-2025	2025	34-36	53	11	31	40	20630	15832	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-27.pdf	
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10-06-2025	2025	34-36	53	11	31	35	27200	22687	https://insp.cd/wp-content/uploads/2025/10/SITREP-MVE-NUM-31.pdf	
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10-14-2025	2025	34-36	53	11	34	4.3	37240	32050	https://insp.cd/wp-content/uploads/2025/10/SITREP39_MVE16.pdf	
10-15-2025	2025	34-36	53	11	34	4.3	37240	32143	https://insp.cd/wp-content/uploads/2025/10/SITREP40_MVE16-B.pdf	

[Screenshot_of_collected_data (Ebola_Congo_2025_16-th_epidemic]

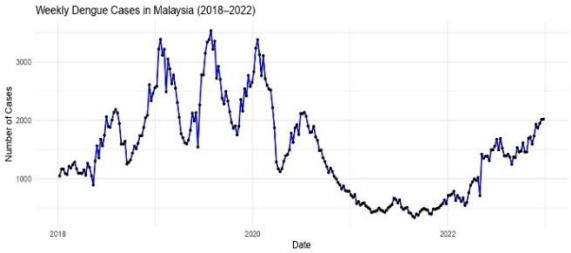
4.2 Dengue Case Data Analysis — Multi-Country (R / RStudio)

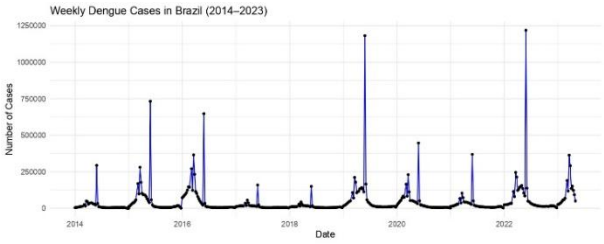
Dengue case data was collected for ten locations: Bangladesh, Brazil, Colombia, Hong Kong, Malaysia, Mexico, Peru, the Philippines, Singapore, and Thailand. For each location, monthly dengue case counts across multiple years were compiled into CSV files and imported into R for processing and visualization.

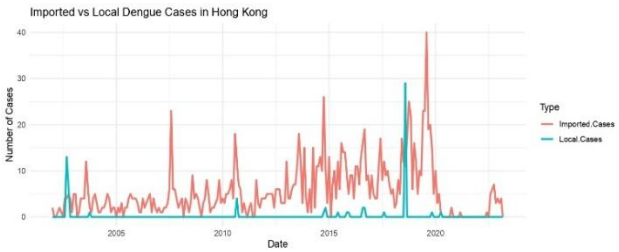
The general workflow followed for each country was:

- Set up the working environment by installing base R followed by RStudio.
- Import raw case-count data (CSV format) using the readr package.
- Reshape the data from wide to long format using tidyr, so that each row represented a single month-year-case count observation.
- Clean and factor the Month variable to ensure correct chronological ordering (January–December).
- Plot case counts using ggplot2, with date/month on the x-axis and number of cases on the y-axis, with separate lines coloured by year.
- Export each figure as a PDF using ggsave() for inclusion in reports.

Some of the plots



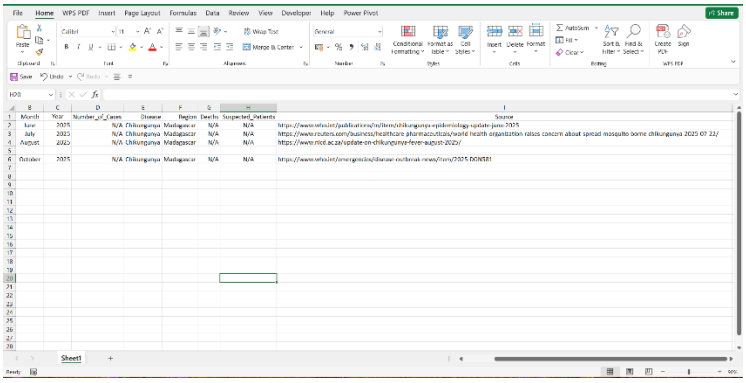




This process was repeated consistently across all ten countries to allow for comparison of seasonal dengue trends between regions.

4.3 Chikungunya Case Data Collection — Madagascar (2025)

Chikungunya case data for Madagascar for the year 2025 was collected and compiled as a supporting dataset for further analysis, following a similar data collection approach to the Ebola and dengue tasks.



Month	Year	Number_of_Cases	Disease	Region	Deaths	Suspected_Patients	Source
June	2025	N/A	Chikungunya	Madagascar	N/A	N/A	https://www.who.int/publications/m/item/chikungunya-epidemiology-update-june-2025
July	2025	N/A	Chikungunya	Madagascar	N/A	N/A	https://www.reuters.com/business/healthcare-pharmaceuticals/world-health-organization-raises-concern-about-spread-mosquito-borne-chikungunya-2025-07-22/
August	2025	N/A	Chikungunya	Madagascar	N/A	N/A	https://www.nicd.ac.za/update-on-chikungunya-fever-august-2025/
October	2025	N/A	Chikungunya	Madagascar	N/A	N/A	https://www.who.int/emergencies/disease-outbreak-news/item/2025-DON581

Data Issues: The dataset contains substantial missing values, an incomplete record, limited observations, and a gap in the monthly data. Important variables such as cases, deaths, and suspected patients are unavailable, requiring data cleaning and further data collection before analysis.

5. Results and Discussion

5.1 Summary of Dengue Case Data Across Ten Locations

Table 1: shows us the dengue case data for each of the ten locations. The data for dengue cases was. Then plotted. The way the data was reported was different for each country. For Bangladesh, Hong Kong and Thailand the dengue case data was reported every month. For Brazil, Colombia, Mexico, Peru, Malaysia, Philippines and Singapore the dengue case data was reported by the week that the epidemiological people use. The time period that the data covered was also different. For Malaysia and Philippines the dengue case data was collected over 5 years. For Bangladesh the dengue case data was collected, over 24 years.

Country	PeriodCovered	Years	TotalCases	Avg. per Interval	Peak Cases	Peak Date	Peak Season
Bangladesh	2000-2023	24	563,674	1,957	79,598	Sep 2023	August
Brazil	2014-2023	10	10,538,911	21,596	1,218,006	28 May 2022	May
Colombia	2014-2023	10	751,564	1,525	5,001	15 Jan 2020	January
Hongkong	2002-2023	22	1,465	5.7	40	Aug 2018	August
Mexico	2014-2023	10	1,155,698	2,340	17,963	15 Oct 2019	October
Malaysia	2018-2022	5	392,839	1,511	3,540	29 Jul 2019	July
Peru	2014-2023	10	623,080	1,272	20,539	22 May 2021	May
Philippines	2018-2022	5	900,753	3,464	23,508	5 Aug 2019	August
Singapore	2012-2023	12	171,221	275	1,791	27 Jul 2020	July
Thailand	2010-2023	14	441,371	3,344	20,978	Aug 2023	July

Table 1. Summary of dengue case statistics by country, computed from the raw datasets used in the R analysis. "Avg. per Interval" refers to the mean cases per reporting interval (monthly or weekly, depending on the country).

5.2 Seasonal Patterns

A clear seasonal signal was observed in every country. Peaks consistently fell within each region's warm/rainy season, when Aedes mosquito breeding conditions are most favourable:

- South/Southeast Asia (Bangladesh, Thailand, Philippines, Malaysia, Singapore): peaks concentrated in July–August, coinciding with the monsoon season.

- Latin America (Brazil, Colombia, Peru, Mexico): peaks spread more widely across January–May and October, reflecting differences in regional rainy-season timing between Southern Hemisphere countries (Brazil, Peru) and countries nearer the equator (Colombia, Mexico).
- Hong Kong showed a much smaller and less consistent seasonal signal, likely because dengue there is overwhelmingly travel-imported rather than locally transmitted, keeping absolute case counts very low (total of 1,465 cases across 22 years, versus over 10 million in Brazil for a comparable period).

5.3 Notable Outbreak Years

Several countries showed a small number of years responsible for a disproportionate share of total cases, consistent with dengue's well-documented multi-year epidemic cycling (driven partly by serotype switching and population immunity):

- Brazil recorded an exceptional epidemic in 2022, with a single week peaking at over 1.2 million reported cases — by far the largest single value in the entire ten-country dataset.
- Bangladesh experienced its most severe outbreak on record in 2023, with monthly cases reaching nearly 80,000 in September, dwarfing every other year in its 24-year series.
- Peru and Thailand both show a similar late-series spike in 2023, suggesting a possible regional or global uptick in dengue transmission that year, consistent with WHO reporting of record global dengue activity in 2023.

5.4 Data Quality Observations

A few data quality issues were identified and addressed during the analysis:

- The Hong Kong dataset contained a mislabelled row (“Total” in place of a month name for early 2023), which was excluded before plotting to avoid a date-parsing error.
- Reporting resolution differs across countries (monthly vs. weekly), which limits direct like-for-like comparison of peak magnitudes and means comparisons across countries should be treated as indicative of relative scale and timing rather than exact equivalence.
- For countries using epidemiological week numbers (Malaysia, Peru, Philippines, Singapore), week-to-calendar-date conversion in R used an approximate method (%U week-of-year); this is adequate for visualizing seasonal trends but is not an exact ISO-8601 epiweek conversion.

5.5 Relation to Objectives

These results directly fulfil the task set: for each of the ten locations, monthly or weekly dengue case counts were plotted with date on the x-axis and number of cases on the y-axis using R and RStudio, allowing seasonal and inter-annual patterns to be visually compared across countries.

6. Challenges Faced

- Inconsistent or incomplete public data sources
- Learning R/RStudio and data-reshaping workflows from scratch
- Reconciling data formats across multiple countries

7. Skills Gained

- Practical experience with R, RStudio, and the tidyverse (tidyr, dplyr, ggplot2, readr)
- Epidemiological data collection and organization
- Data visualization and time-series plotting
- Working independently under academic research guidance
- For Data storage, file organization, and sharing, I learned a new tool called Dropbox

8. Conclusion

This online internship provided valuable practical experience in the collection and analysis of infectious disease surveillance data across three distinct disease contexts—Ebola, dengue, and chikungunya. The work strengthened my technical skills in R-based data analysis and broadened my understanding of epidemiological data practices through online mentorship and guidance from a respected Professor at a university in Hong Kong.

9. References

List data sources :

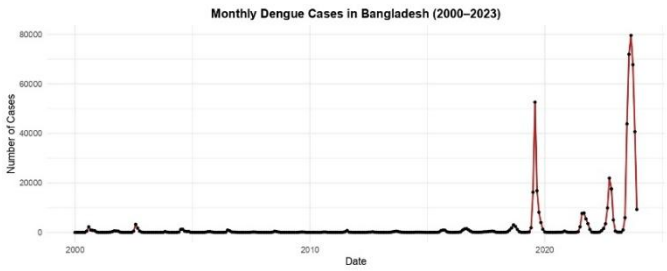
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- https://insp.cd/wp-content/uploads/2025/09/SITREP-MVE-12.pdf
- Others…..

Appendix

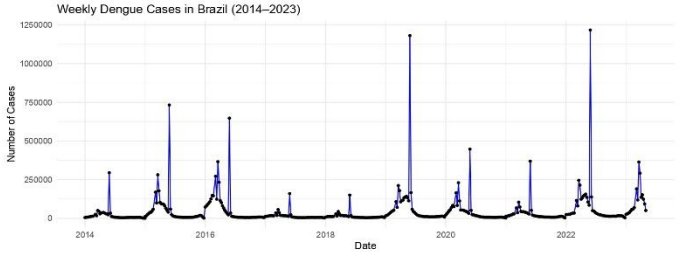
Country

Plots

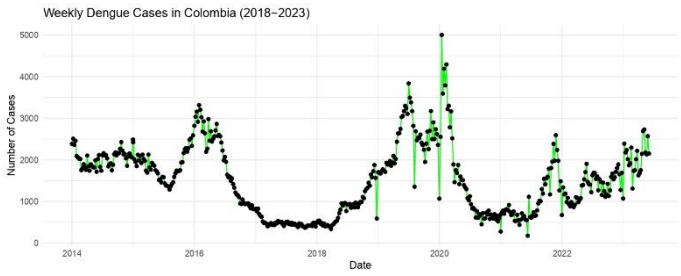
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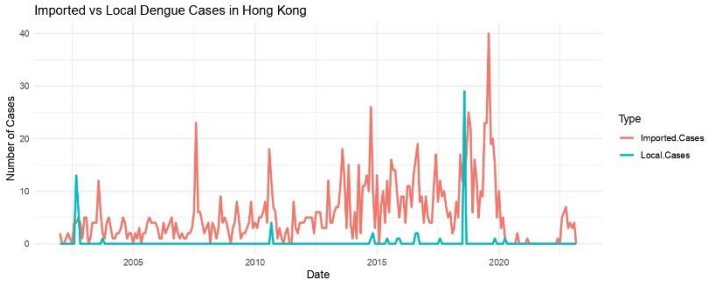
2. Brazil_Dengue_cases



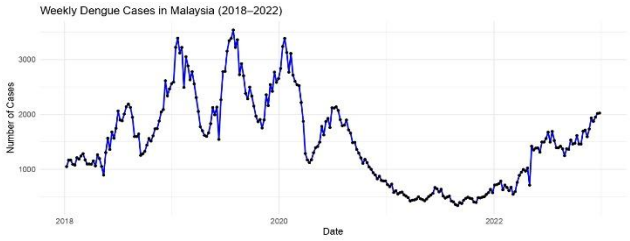
3. Colombia_Dengue_cases



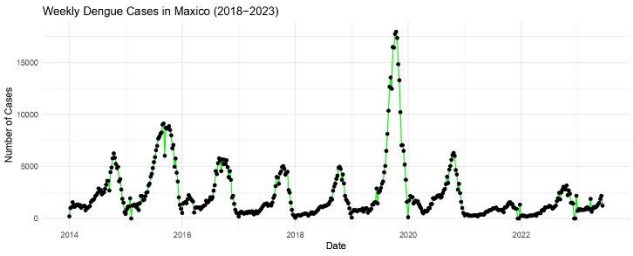
4. Hongkong_Dengue_cases



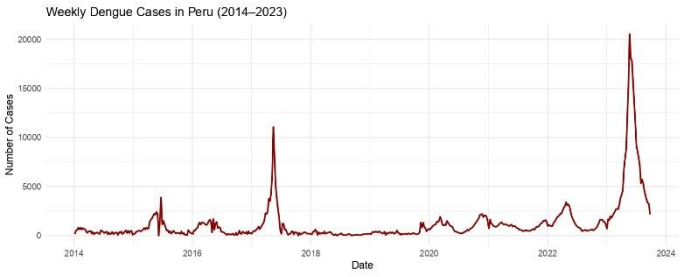
5. Malaysia_Dengue_cases



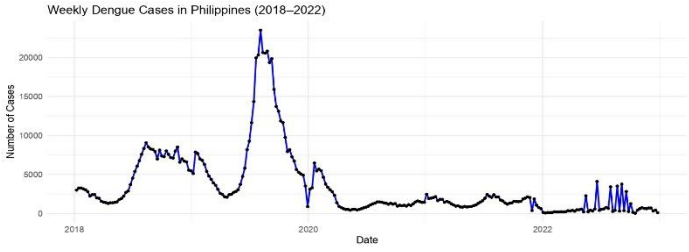
6. Mexico_Dengue_cases



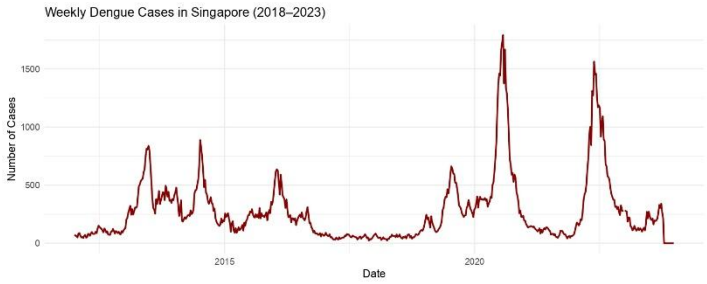
7. Peru_Dengue_cases



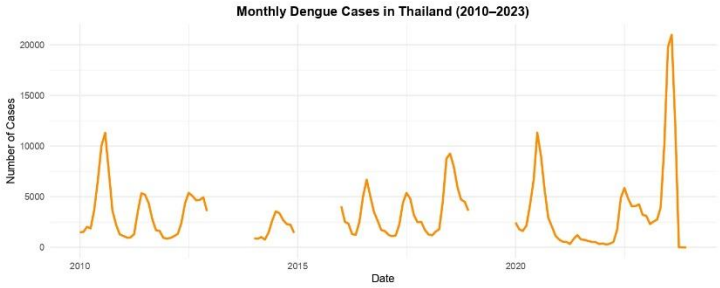
8. Philippines_Dengue_cases



9. Singapore_Dengue_cases



10. -Thailand_Dengue_cases



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